

Learn Prolog Now!

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7.3 Exercises

Exercise 7.1 Suppose we are working with the following DCG:

```
s --> foo,bar,wiggle.
foo --> [choo].
foo --> foo,foo.
bar --> mar,zar.
mar --> me,my.
me --> [i].
my --> [am].
zar --> blar,car.
blar --> [a].
car --> [train].
wiggle --> [toot].
wiggle --> wiggle,wiggle.
```

Write down the ordinary Prolog rules that correspond to these DCG rules. What are the first three responses that Prolog gives to the query `s(X,[])` ?

Exercise 7.2 The formal language $a^n b^n - \{ \epsilon \}$ consists of all the strings in $a^n b^n$ except the empty string. Write a DCG that generates this language.

Exercise 7.3 Let $a^n b^{2n}$ be the formal language which contains all strings of the following form: an unbroken block of `a` s of length `n` followed by an unbroken block of `b` s of length `2n` , and nothing else. For example, `abb` , `aabbbb` , and `aaabbbbbbb` belong to $a^n b^{2n}$, and so does the empty string. Write a DCG that generates this language.

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